DIPOLE TYPE BY LENGTH
l = antenna rod length (m)

$$\boxed{\lambda \text{ (wavelength)} = \frac{c}{f} = \frac{3\times10^8}{f} \text{ (m)}}$$

l/λ	Type	Use formula
$< 1/50$	Hertzian (short)	HERTZIAN $S(R,\theta)$, $D=1.5$
$= 1/4$	Quarter-wave	ARB formula, $\pi l/\lambda = \pi/4$
$= 1/2$	Half-wave	half-wave shortcut, $D=1.64$, $R_{\text{rad}}=73\,\Omega$
$= 1$	Full-wave	ARB, $D=2.41$, $R_{\text{rad}}=199\,\Omega$
other	Arbitrary	ARB formula

HERTZIAN DIPOLE — $S(R,\theta)$

Far-field E,H phasors *small $l \ll \lambda$*

$$\tilde{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi}\,\frac{e^{-jkR}}{R}\sin\theta, \quad \tilde{H}_\phi = \tilde{E}_\theta/\eta_0$$

Avg power density *$l/\lambda < 1/50$; far field*

$$\boxed{S(R,\theta) = \frac{\eta_0\, k^2 I_0^2\, l^2}{32\pi^2 R^2}\sin^2\theta} \text{ (W/m}^2\text{)}$$

Equivalent: $S = S_{\max}\sin^2\theta$ *with* $S_{\max} = \eta_0 k^2 I_0^2 l^2/(32\pi^2 R^2)$ *at* $\theta = \pi/2$.

l: rod length (m)	I_0: peak current (A)
R: obs. distance (m)	θ: angle from dipole axis (rad)
$k = 2\pi/\lambda$ (rad/m)	$\eta_0 = 120\pi \approx 377\,\Omega$

Total radiated power $P_{\text{rad}} = \frac{\eta_0\pi}{3}(I_0 l/\lambda)^2$ (W)

ARBITRARY-LENGTH DIPOLE — $S(\theta)$

Avg power density *any l; far field*

$$\boxed{S(R,\theta) = \frac{15 I_0^2}{\pi R^2}\left[\frac{\cos\left(\frac{\pi l}{\lambda}\cos\theta\right) - \cos\left(\frac{\pi l}{\lambda}\right)}{\sin\theta}\right]^2} \text{ (W/m}^2\text{)}$$

At $\theta = 0, \pi$: bracket is 0/0. L'Hôpital $\Rightarrow$ $S = 0$ (no axial radiation). TI-36X alt: evaluate bracket at $\theta = 0.001$ rad $\to$ tiny $\to$ 0.

Normalized pattern *dimensionless* $F(\theta) = S(\theta)/S_{\max}$

COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

Conversion *multiply by $\pi/180$ or $180/\pi$*

$$\theta_{\text{rad}} = \theta_{\text{deg}}\cdot\frac{\pi}{180} \qquad \theta_{\text{deg}} = \theta_{\text{rad}}\cdot\frac{180}{\pi}$$

°	rad	$\sin\theta$	$\cos\theta$	$\sin^2\theta$	$\cos^2\theta$
0	0	0	1	0	1
30	$\pi/6$	$1/2$	$\sqrt{3}/2$	$1/4$	$3/4$
45	$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$	$1/2$	$1/2$
60	$\pi/3$	$\sqrt{3}/2$	$1/2$	$3/4$	$1/4$
90	$\pi/2$	1	0	1	0
180	π	0	-1	0	1

Antenna formulas usually take θ in rad. RAD mode on calc.

LINEAR ANTENNA ARRAY
N = # of elements, d = interelement spacing (m)

Array factor (uniform, broadside) *$k = 2\pi/\lambda$, $\gamma = kd\cos\theta$*

$$F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}; \quad F_a^{\max} = N^2 \text{ at } \theta = \pi/2$$

Normalize $F_a^{\text{norm}} = F_a/N^2$

HPBW (broadside, uniform) *use Nd, not $(N-1)d$*

$$\beta \approx 0.886\,\frac{\lambda}{Nd} \text{ (rad)} = 50.8^\circ\cdot\frac{\lambda}{Nd}$$

Electronic steering (scan to θ_0) *replace γ with $\gamma' = kd(\cos\theta - \cos\theta_0)$*

$$\delta = \frac{2\pi d}{\lambda}\cos\theta_0\ \left(\frac{\text{rad}}{\text{elt}}\right); \quad \theta_0 = \frac{\pi}{2}\text{:broadside, 0:end-fire}$$

Frequency scan *feed-line incr. $\Delta\ell = n_0\lambda_0$*

$$\cos\theta_0 \approx \frac{n_0\lambda_0}{d}\,\frac{\Delta f}{f_0}$$

Grating lobes appear if $d > \lambda$. Avoid.

BEAM PARAMETERS — β, Ω_p, D

Normalize *always start here*

$$F(\theta) = S(\theta)/S_{\max} \text{ (unitless)}$$

Beamwidth β at threshold X

Set $F(\theta_X) = X$, solve for θ_X; $\beta = 2\theta_X$ (symmetric)

Threshold	$X = 10^{\text{dB}/10}$	θ_X
HPBW (-3 dB)	0.5	$\sqrt{\frac{\ln 2}{a}}$
-6 dB	0.25	$\sqrt{\frac{\ln 4}{a}}$
-10 dB	0.1	$\sqrt{\frac{\ln 10}{a}}$
any $F(\theta_X)$	any X	$F(\theta_X) = X$

Pattern solid angle Ω_p *recognize pattern, look up*

$$\Omega_p = \int_0^{2\pi}\int_0^{\pi} F(\theta)\sin\theta\, d\theta\, d\phi \text{ (sr)}$$

No ϕ dep.: $\Omega_p = 2\pi\int_0^\pi F(\theta)\sin\theta\, d\theta$.

θ is the integration variable — DON'T plug in a number for θ (e.g. $\theta_{1/2}$). Integrate over $\theta \in [0,\pi]$.

Pattern $F(\theta)$	Ω_p (sr)
Isotropic ($F = 1$)	$4\pi \approx 12.57$
Hertzian $\sin^2\theta$	$8\pi/3 \approx 8.38$
Half-wave dipole	≈ 7.66
Narrow Gauss $e^{-a\theta^2}$	$\pi/a = \pi\theta_{1/2}^2/\ln 2$
2-axis HPBW (aperture)	$\beta_{xz}\cdot\beta_{yz}$
Other (use integral)	numerical

TI-36X: `[2nd][d/dx]` for $\int_\square^\square$, RAD mode, multiply by 2π.

Directivity D *always $D = 4\pi/\Omega_p$; common shortcuts below*

$$D = \frac{4\pi}{\Omega_p} \text{ (unitless)} \qquad D_{\text{dB}} = 10\log_{10} D$$

WORD TRAP: "direction" = $\theta_{\max}$ (rad); "directivity" = D (unitless).

Gain (if asked) *ξ = radiation efficiency (unitless), $0 < \xi \le 1$*

$$G = \xi D \text{ (unitless)} \qquad G_{\text{dB}} = 10\log_{10} G \text{ (dB)}$$

Lossless / ideal antenna $\Rightarrow G = D$.

$\xi = \frac{R_{rad}}{R_{rad} + R_{loss}} = \frac{P_{rad}}{P_{rad} + P_{loss}}$; $R_{rad} = 2P_{rad}/I_0^2$ (Ω); $R_{in} = R_{rad} + R_{loss}$.

COMMON DIPOLE PARAMETERS

Lossless ($\xi = 1$): $G = D$.

Type	l	D	D_{dB}	R_{rad} (Ω)
Isotropic	N/A	1	0	—
Hertzian	$< \lambda/50$	1.5	1.76	$80\pi^2(l/\lambda)^2$
Half-wave	$\lambda/2$	1.64	2.15	73.2
Monopole	$\lambda/4$ (gnd)	1.64	2.15	36.6
Full-wave	λ	2.41	3.82	199

Half-wave: $P_{rad} = 36.6 I_0^2$ W. Monopole ($\lambda/4$ above ground): same patt. as half-wave but P_{rad}, R_{rad} halved.

ANTENNA AREAS (A_e, A_p)

Effective area *antenna's RX cross section; D from BEAM PARAMS or table above*

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}$$

Physical cross section *wire dipole, diameter d; l see table above*

$$A_p = l\,d \text{ (m}^2\text{, any dipole)}$$

Inverse: D from A_e

$$D = \frac{4\pi A_e}{\lambda^2} \text{ (unitless)}$$

Thin wire: $A_e \gg A_p$ (e.g., Hertzian $A_e = 3\lambda^2/8\pi \approx 0.119\lambda^2$).

FRIIS TRANSMISSION FORMULA

For each antenna's G: if given by NAME ("half-wave dipole" etc.) $\to$ COMMON DIPOLE PARAMS ($G = D$, lossless). Else (dB/linear value given) $\to$ use as given (convert dB $\to$ linear).

Pwr density at RX, then RX pwr *G_t, G_r linear gains; $\lambda = c/f$*

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{)}; \quad \boxed{P_r = G_t G_r\left(\frac{\lambda}{4\pi R}\right)^2 P_t} \text{ (W)}$$

Equivalent: $P_r = S_r A_{e,r}$.

Solve for R *range*

$$R = \frac{\lambda}{4\pi}\sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using A_e *recv. area form*

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss: $L_{fs} = (4\pi R/\lambda)^2$, *so* $P_r = G_t G_r P_t/L_{fs}$.

Off-axis (patterns not aligned): multiply by $F_t(\theta_t,\phi_t)F_r(\theta_r,\phi_r)$, both = 1 when aligned.

Recipe 1. $\lambda = c/f$. 2. Convert any dB gain $\to$ linear: $G = 10^{G_{\text{dB}}/10}$.
3. Get G_t, G_r by antenna type: dipole$\to$COMMON DIPOLE PARAMS; aperture$\to$APERTURE box; arbitrary $F \to$BEAM PARAMS. 4. Plug into Friis.

Use linear G, not dB. Convert FIRST.

dB $\leftrightarrow$ LINEAR

Works for any power-like quantity (G, D, P_r/P_t, SNR, . . .):

$$X_{\text{dB}} = 10\log_{10} X_{\text{lin}} \qquad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

Memorize: 3 dB $\Leftrightarrow \times 2$, 10 dB $\Leftrightarrow \times 10$. Add dB = multiply linear.

For E-field / amplitude use $20\log$ not $10\log$ (power $\propto E^2$).

NOISE & SNR

Noise power *thermal noise into matched receiver*

$$P_N = k_B\, T_{\text{sys}}\, B \text{ (W)}$$

$k_B = 1.38\times10^{-23}$ J/K; T_{sys} system noise temp (K); B receiver bandwidth (Hz).

Signal-to-noise ratio

$$\text{SNR} = \frac{P_{\text{rec}}}{P_N} \text{ (unitless)} \qquad \text{SNR}_{\text{dB}} = 10\log_{10}\frac{P_{\text{rec}}}{P_N}$$

APERTURE ANTENNAS (horn, dish)

A_p physical aperture area (m^2); A_e effective area (m^2, $= \lambda^2 D/4\pi$); β_{xz}, β_{yz} HPBWs in two planes (rad); ℓ, d aperture side / diameter (m).

Far-field req.: $R \ge 2d^2/\lambda$ (d = largest aperture dim.).

Pattern (rect., uniform illum.) *peak at $\theta = 0$ (broadside)*

$$F(\theta) = \text{sinc}^2\left(\frac{\pi\ell_x\sin\theta}{\lambda}\right); \quad S_0 = \frac{E_0^2 A_p^2}{2\eta_0\lambda^2 R^2}$$

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \quad \Rightarrow \quad A_e \approx A_p$$

Directivity from beamwidths *any aperture*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

HPBW and D by aperture shape *uniform illum.; β in rad*

Shape	A_p	HPBW (rad)	D (unitless)
Rect. $\ell_x \times \ell_y$	$\ell_x\ell_y$	$\beta_x \approx 0.88\lambda/\ell_x$ $\beta_y \approx 0.88\lambda/\ell_y$	$D \approx 4\pi/(\beta_x\beta_y)$ $= 4\pi\ell_x\ell_y/\lambda^2$
Square ℓ	ℓ^2	$\beta \approx 0.88\lambda/\ell$	$D \approx 4\pi/\beta^2 = 4\pi\ell^2/\lambda^2$
Circular (diam. d)	$\pi d^2/4$	$\beta \approx 1.02\lambda/d$	$D \approx 4\pi/\beta^2 = \pi^2 d^2/\lambda^2$

Scaling (any ratio) *$D \propto A_p/\lambda^2$; $\beta \propto \lambda/\sqrt{A_p}$*

$$D_{\text{new}} = D_{\text{old}}\cdot\frac{A_{p,\text{new}}}{A_{p,\text{old}}}\cdot\left(\frac{f_{\text{new}}}{f_{\text{old}}}\right)^2$$

$$\beta_{\text{new}} = \beta_{\text{old}}\cdot\sqrt{\frac{A_{p,\text{old}}}{A_{p,\text{new}}}}\cdot\frac{f_{\text{old}}}{f_{\text{new}}}$$

If a parameter is unchanged, its ratio = 1. Examples: double area $\to D\times2$, $\beta\times1/\sqrt{2}$; double freq $\to D\times4$, $\beta\times1/2$.

DIPOLE

l — antenna length (m)
$\lambda = c/f$ (m); $c = 3\times10^8$ m/s
I_0 — peak current (A)
$k = 2\pi/\lambda$ (rad/m)
R — obs. dist. (m); θ from axis
$\eta_0 = 120\pi \approx 377\,\Omega$
$S(R,\theta)$ — pwr density (W/m^2)

BEAM

$F(\theta) = S/S_{\max}$ (unitless)
a — coef. on θ^2 in Gaussian
β — HPBW (rad or °)
Ω_p — pattern solid angle (sr)
$D = 4\pi/\Omega_p$ — directivity
ξ radiation eff. (unitless); $G = \xi D$
$D_{\text{dB}} = 10\log_{10} D$; $D = 10^{D_{\text{dB}}/10}$

FRIIS / dB

P_t — power transmitted (W)
P_r — power received (W)
G_t — transmit gain (linear)
G_r — receive gain (linear)
$S_r = G_t P_t/(4\pi R^2)$ (W/m^2)
$P_r = G_t G_r(\lambda/4\pi R)^2 P_t$
$G = 10^{G_{\text{dB}}/10}$
3 dB $= \times2$, 10 dB $= \times10$
$P_N = k_B T_{\text{sys}} B$ (W)

APERTURE

A_p — physical area (m^2)
$A_e = \lambda^2 D/(4\pi)$ (m^2)
$A_e \approx A_p$ for aperture
$D \approx 4\pi A_p/\lambda^2$ (unitless)
$\beta \approx 0.88\lambda/\ell$ (rect/sq.)
$\beta \approx 1.02\lambda/d$ (circular)
$2A_p \Rightarrow 2D$, $\beta/\sqrt{2}$

ARRAY

N — number of elements
d — spacing (m)
$\gamma = kd\cos\theta$
$F_a = \sin^2(N\gamma/2)/\sin^2(\gamma/2)$
$F_a^{\max} = N^2$ at $\theta = \pi/2$
$F_a^{\text{norm}} = F_a/N^2$
$\beta \approx 0.886\lambda/(Nd)$ (broadside)